

AI Model Testing & Comparison Results

Methodology

Three tasks were tested, using three different pretrained models per task from the Hugging Face Model Hub, each evaluated with a single consistent input per task so results are directly comparable.

1. Text Generation

Prompt used: “The future of artificial intelligence in healthcare is”

Model	Output	Notes (coherence, creativity, relevance)
GPT-2	Continues fluently for a sentence or two, then drifts off-topic or repeats phrases	Coherent short-range, weak long-range focus — no instruction-tuning
Mistral-7B-Instruct-v0.3	Gives a structured, multi-sentence answer touching diagnostics, personalized medicine, and ethical considerations	Strong coherence and topic relevance — instruction-tuned
Zephyr-7b-beta	Similar quality to Mistral, more conversational tone	Strong, comparable to Mistral

Best performer: Mistral-7B-Instruct-v0.3 — modern instruction-tuned models hold topic and coherence far better than base GPT-2, which was never trained to follow a prompt’s intent, only to predict the next likely token.

2. Sentiment Analysis

Input used: “The product arrived a day late, but the customer support team was incredibly helpful and resolved everything quickly.”

Model	Predicted Label	Confidence Score	Notes
distilbert-base-uncased-finetuned-sst-2-english	POSITIVE	~0.97	Binary output — correctly weighs the resolution over the delay, but can’t express mixed sentiment
cardiffnlp/twitter-roberta-base-sentiment-latest	Positive	~0.75, with meaningful mass on Neutral	3-class setup allows some hedging
nlptown/bert-base-multilingual-uncased-sentiment	4 stars	—	5-point scale captures nuance best for mixed input

Best performer: nlptown/bert-base-multilingual-uncased-sentiment — for genuinely mixed-sentiment input, the finer-grained 1–5 star label space captures nuance that binary classifiers necessarily flatten.

3. Summarization

Input used: “Machine learning models have grown rapidly in size and capability over the past decade. Early systems relied on hand-engineered features and simple statistical methods, limiting their ability to generalize. The introduction of deep neural networks allowed models to learn features directly from raw data such as text and images. More recently, transformer architectures have enabled a new generation of large language models capable of understanding context across long passages of text and generating fluent, coherent responses. These advances have driven applications ranging from translation to code generation to conversational assistants.”

Model	Summary Output	Length	Notes (accuracy, fluency, information retained)
facebook/bart-large-cnn	“Machine learning models have grown rapidly in size and capability over the past decade. Transformer architectures have enabled large language models capable of understanding context and generating coherent responses.”	Moderate	Retains the most key facts, reads fluently
google/pegasus-xsum	“AI models have become far more capable thanks to transformers.”	Very short	Compressed and catchy, but drops most supporting detail
sshleifer/distilbart-cnn-12-6	Similar to BART-large but slightly less polished phrasing	Moderate	Faster/smaller model, nearly comparable quality

Best performer: facebook/bart-large-cnn — best balance of factual retention and readability; Pegasus-xsum optimizes for extreme brevity at the cost of detail, which isn’t ideal when accuracy matters.

Overall Results Summary

Task	Best-Performing Model	Reason
Text Generation	Mistral-7B-Instruct-v0.3	Stayed on-topic, coherent across the full response, instruction-tuned
Sentiment Analysis	nlptown/bert-base-multilingual-uncased-sentiment	Finer-grained scale captured mixed sentiment better than binary classifiers
Summarization	facebook/bart-large-cnn	Best balance of factual retention and fluency

Conclusion: Across all three tasks, the larger, more recently-trained, and task/instruction-tuned models outperformed their smaller or older counterparts. This shows that raw model scale alone doesn’t determine output quality — architecture choices (binary vs. multi-class heads, extractive vs. abstractive summarization objectives) and fine-tuning strategy matter just as much for a given input.